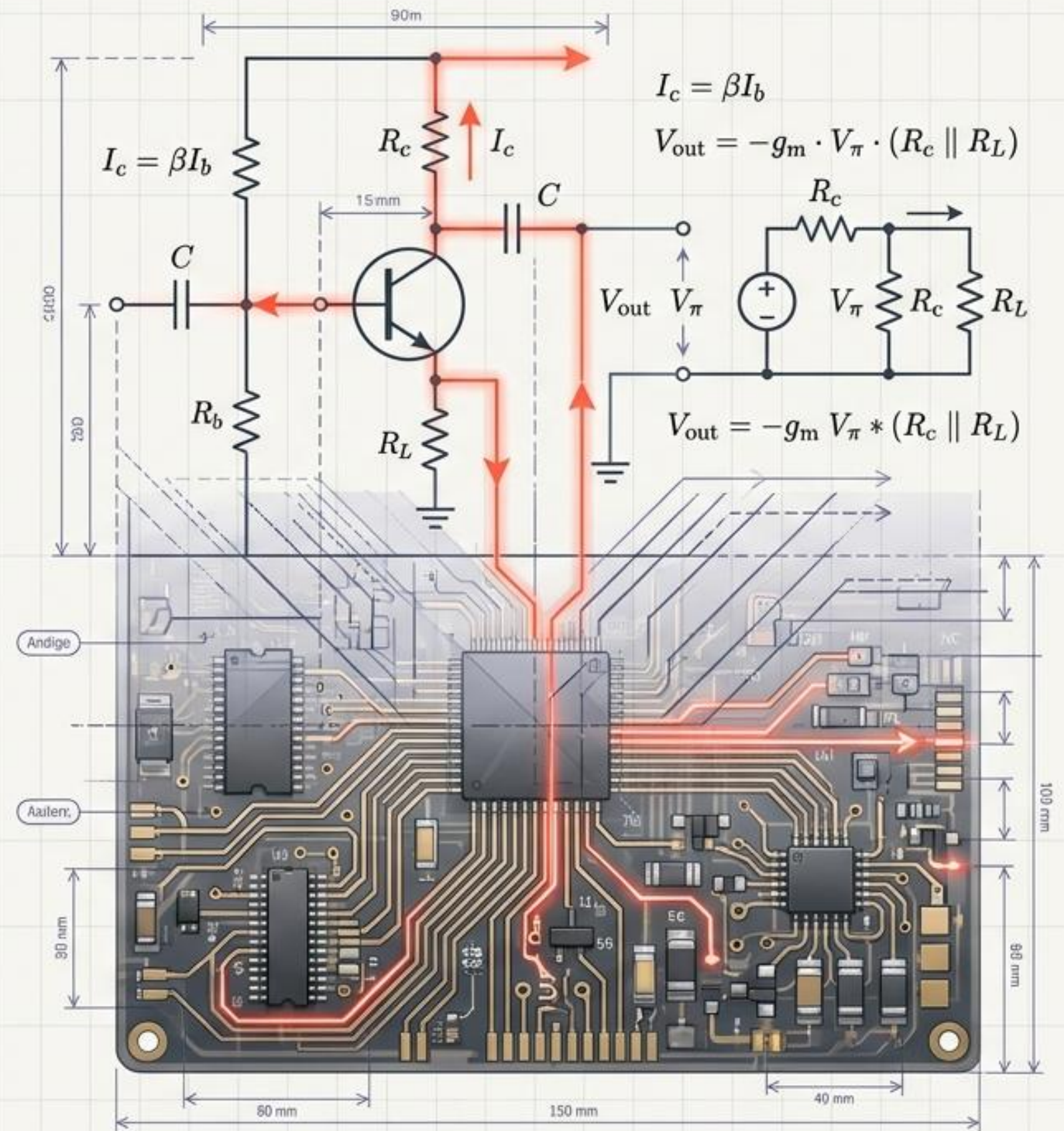


BJT AC Analysis

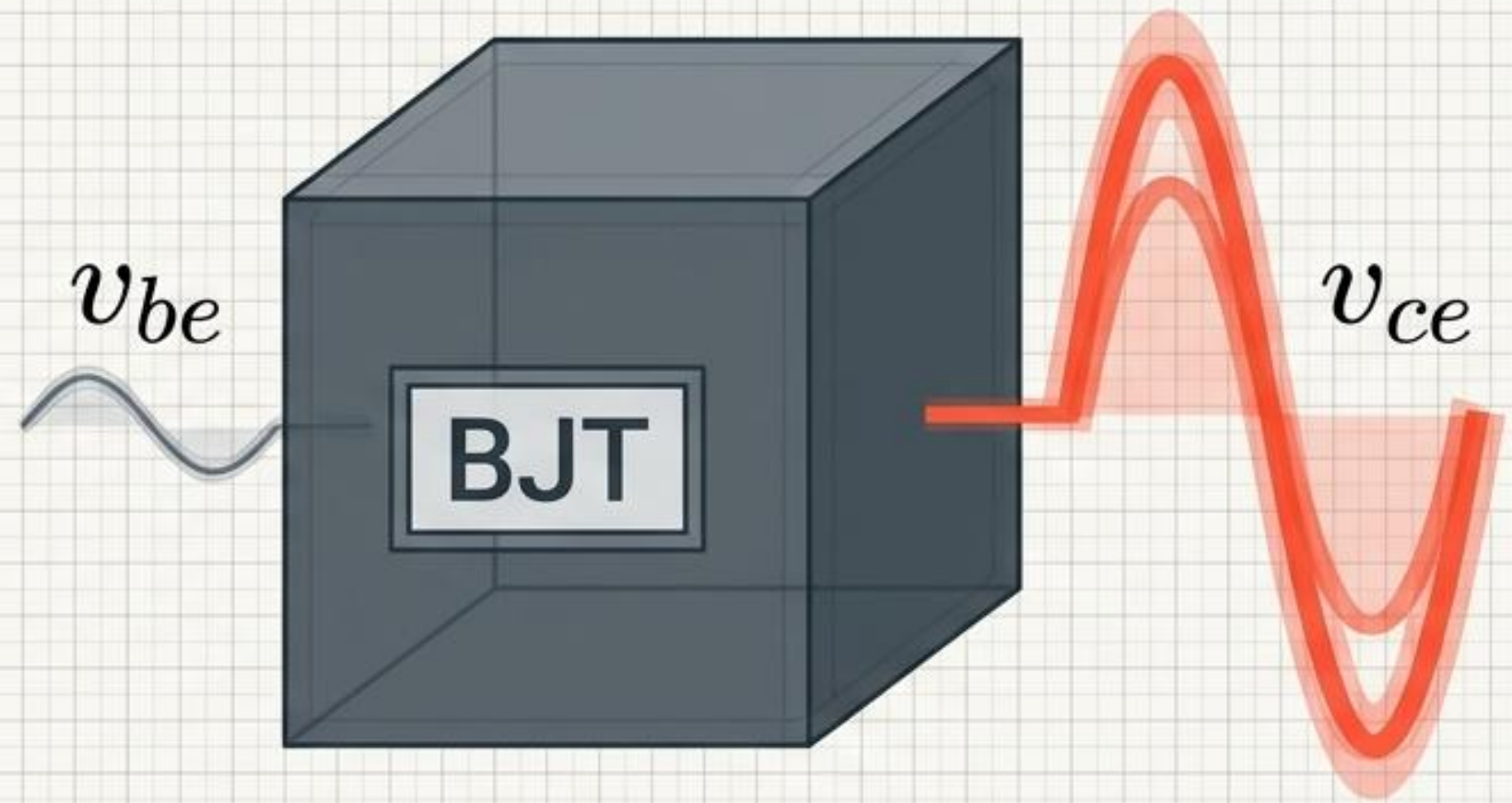
Deconstructing Small-Signal Amplification and Transistor Modeling



The Foundation of Modern Electronics

The invention of amplifying devices—initially vacuum tubes, later transistors—ignited the evolution of modern technology.

Without amplification, radio, television, and computing would not exist.



A microscopic input signal injected into the base produces an identical, massively magnified collector voltage.

Amplifying at the Q-Point

Input Signal (v_{be})

A small signal voltage oscillates around the base.

The Q-Point

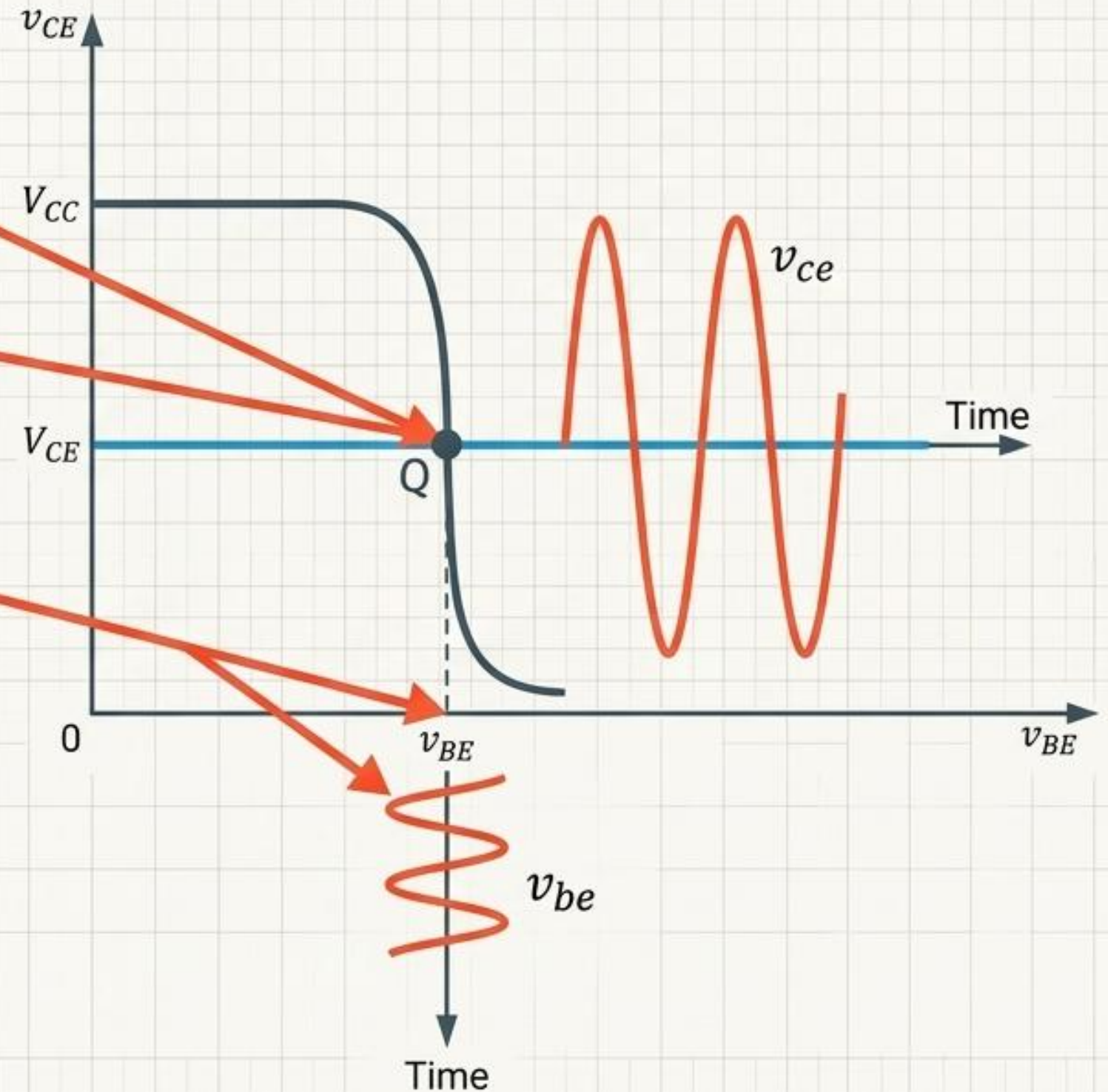
The transistor must be biased near the middle of the load line to allow full, unclipped swing.

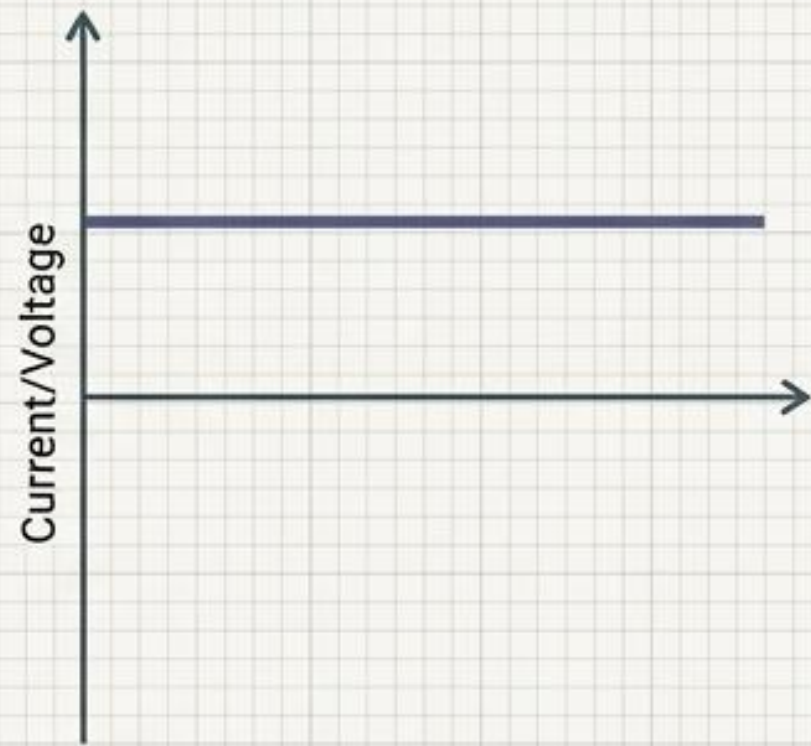
Output Signal (v_{ce})

The output appears superimposed on the DC collector voltage (V_{CE}).

$$\text{Voltage Gain } (A_v) = -\frac{v_{ce}}{v_{be}}$$

The amplitude of the output is larger than the input by this exact multiplier.

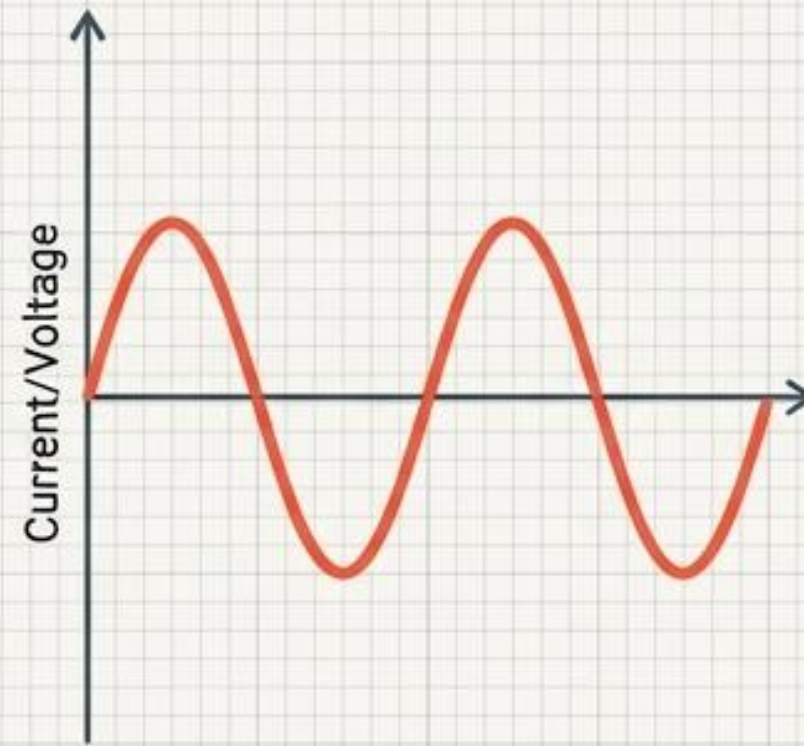




The DC Foundation

Example: $I_C = 2.93 \text{ mA}$
 $V_C = 15.35 \text{ V}$

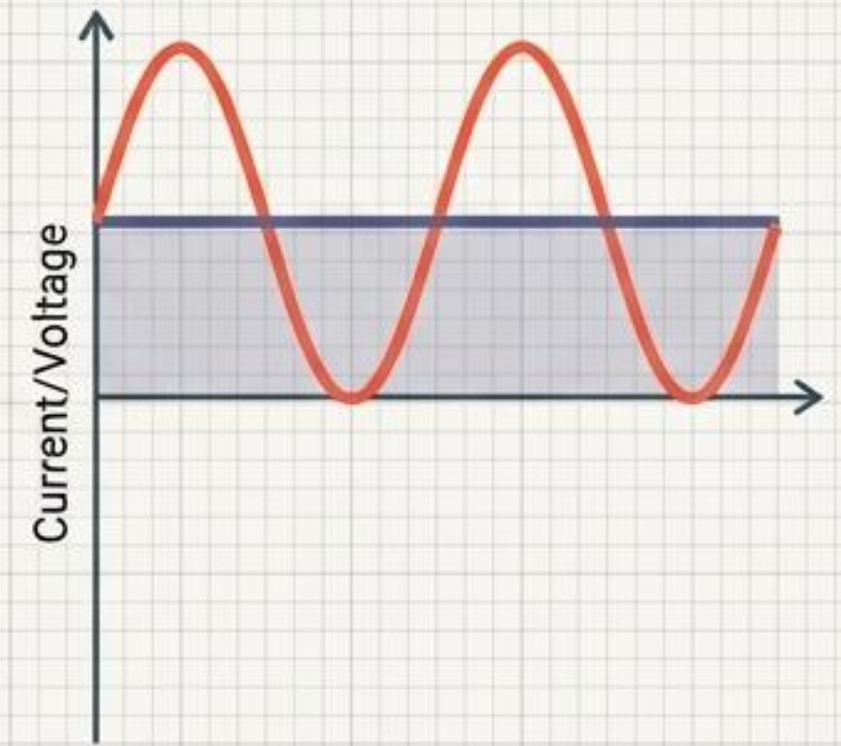
+



The Dynamic Input

i_c (AC only)

=



The Superimposed Output

The total signal is the precise combination of the fixed bias and the fluctuating waveform.

$$\text{Total } i_C = DC + AC$$

The Engineer's Lexicon

CAPITAL LETTERS

Reserved exclusively for static DC quantities.

Examples: I_C (Collector Current),
 V_{CE} (Collector-Emitter Voltage),
 I_B (Base Current)

Gain: $\beta_{DC} = \frac{I_C}{I_B}$ (DC Current Gain)

lowercase letters

Reserved exclusively for dynamic AC quantities.

Examples: i_c (ac collector current),
 v_{ce} (ac collector-emitter voltage),
 i_b (ac base current)

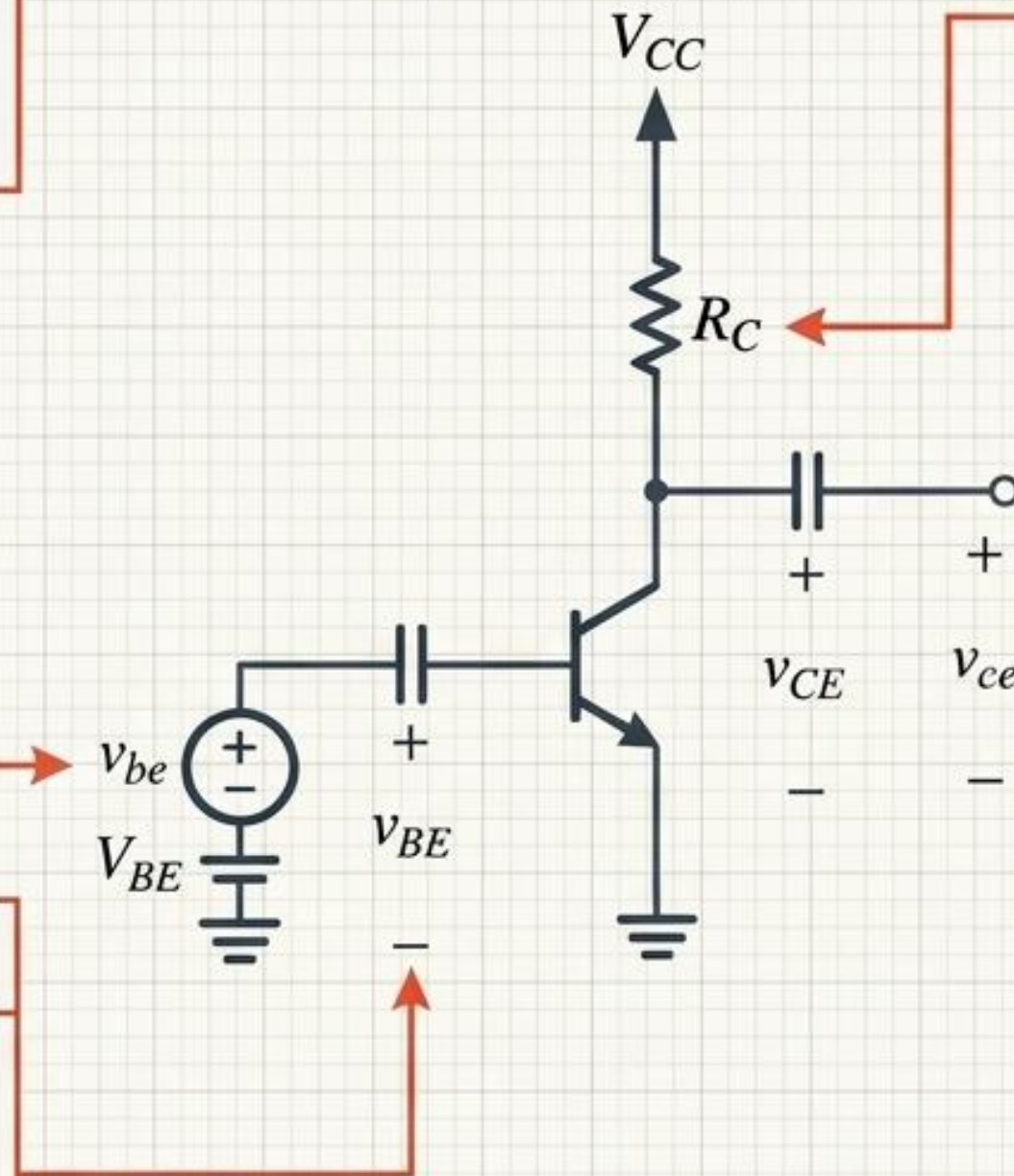
Gain: $\beta_{ac} = \frac{i_c}{i_b}$ (AC Current Gain)

The Input

A $100\mu\text{V}$ AC source (v_{be}) injected into the base.

The Bias

V_{CC} (+30V) driving the DC foundation through R_C ($5\text{k}\Omega$).



The Gates

Coupling capacitors at the input and output to isolate the AC signal from DC voltage sources.

The Output

The amplified, inverted signal (v_{ce}) exiting the collector.

DC State

- Condition:
Frequency (f) = 0
- Math Result:
Reactance (X_C)
becomes infinitely large.



Circuit Behavior: acts
as an OPEN circuit.
Current (i_C) = 0.

$$\text{Reactance } X_C = \frac{1}{2\pi fC}$$

AC State

- Condition:
Frequency (f) is large
- Math Result:
Reactance (X_C)
approaches 0.

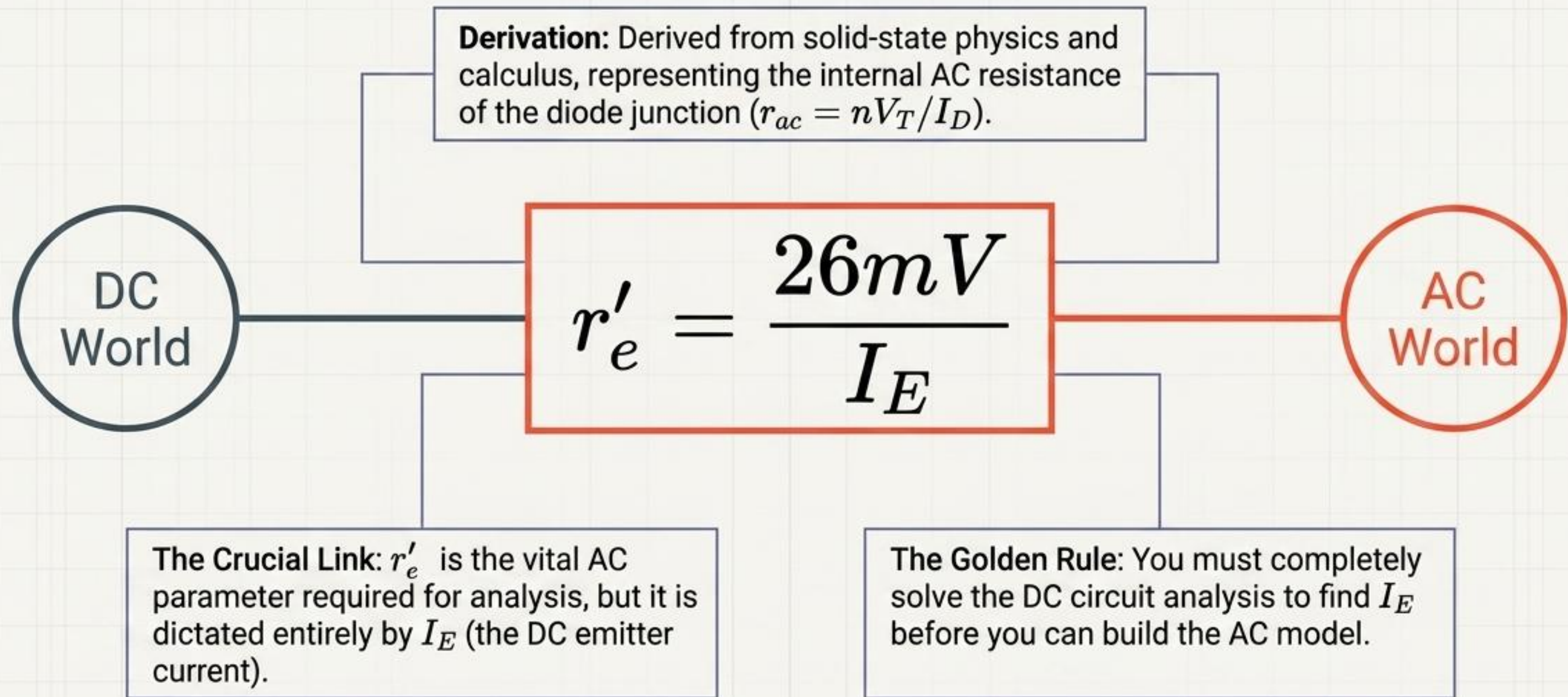


Circuit Behavior: acts
as a SHORT circuit.
Current passes freely.

Capacitor Deployment: Coupling vs. Bypass

	Location	DC Role	AC Role
Coupling Capacitors	Positioned between the BJT and the input/output stages.	Blocks DC. Prevents the source and load from disrupting the carefully established DC bias.	Passes the AC signal directly into and out of the amplifier stage.
Bypass Capacitors	Positioned parallel to the emitter resistor.	Necessary for DC stability (acts as an open circuit, leaving the emitter resistor active).	Shorts the emitter directly to ground for AC signals. Ensures AC signal appears entirely across base-emitter junction for higher voltage gain.

Bridging the Worlds: AC Emitter Resistance



Superposition Theorem in Practice

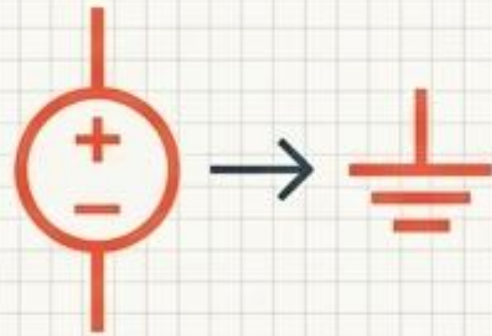
To analyze the AC performance, we must mathematically isolate the AC components by stripping away the DC foundation.

1

Neutralize DC Sources

Action: Set all DC voltage and current sources to zero.

Visual Rule: Replace V_{CC} lines with solid ground connections mentally.

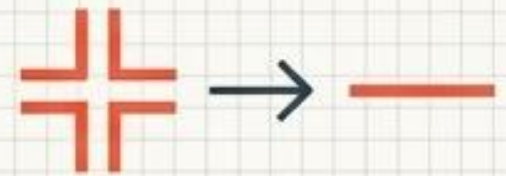


2

Short the Gatekeepers

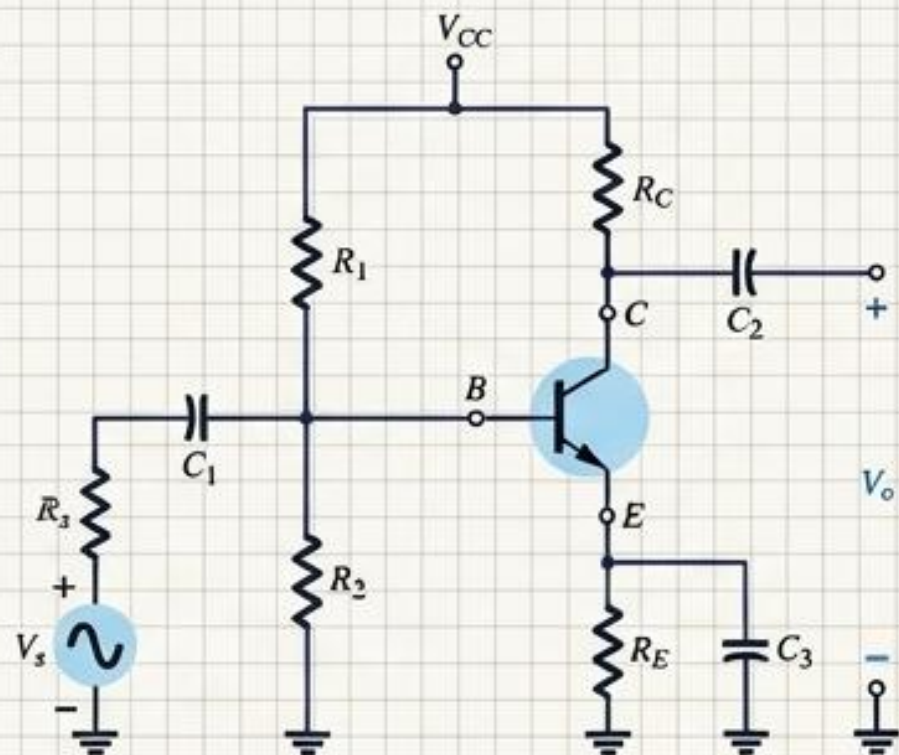
Action: Treat all large capacitors as ideal pathways for high-frequency AC.

Visual Rule: Replace coupling and bypass capacitors with solid short-circuit lines.



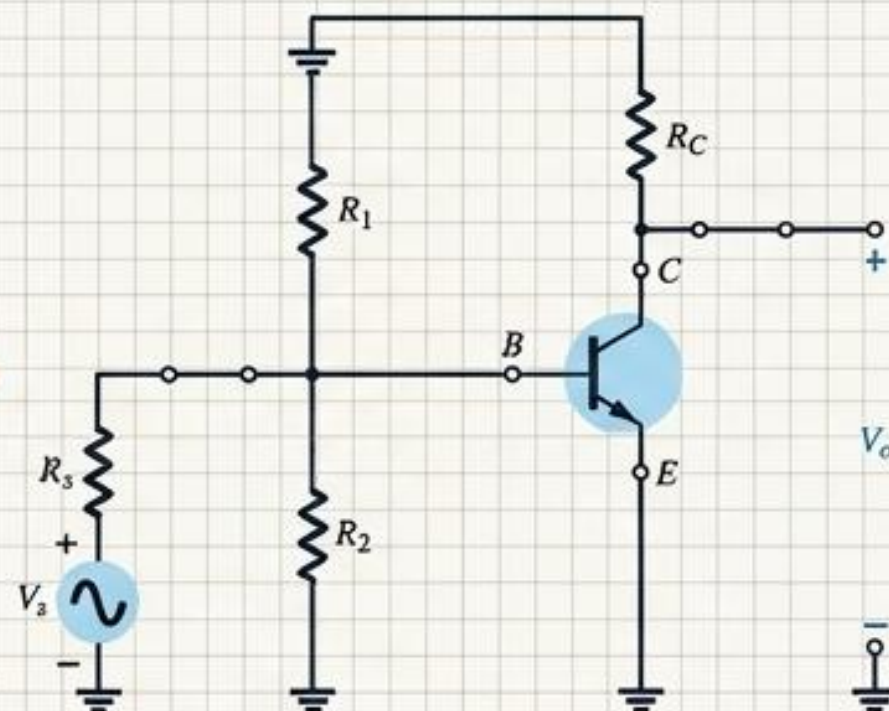
Revealing the AC Equivalent Circuit

The Physical Reality



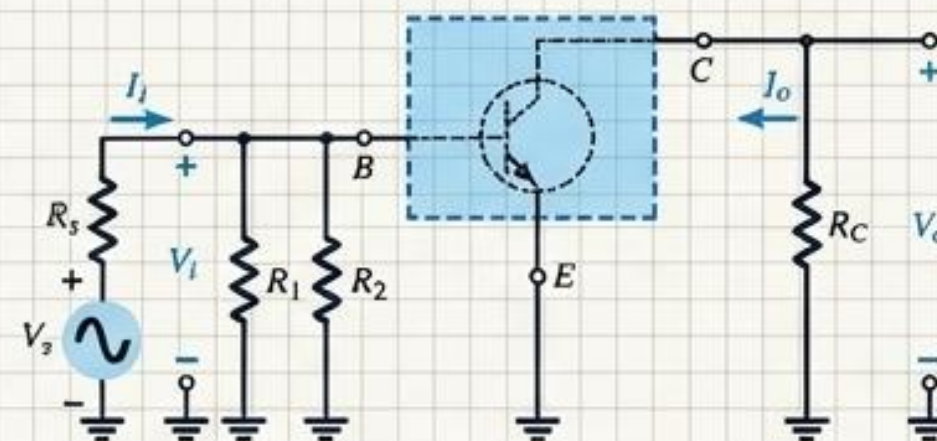
The complete BJT network with mixed DC sources and physical capacitors.

Applying Superposition



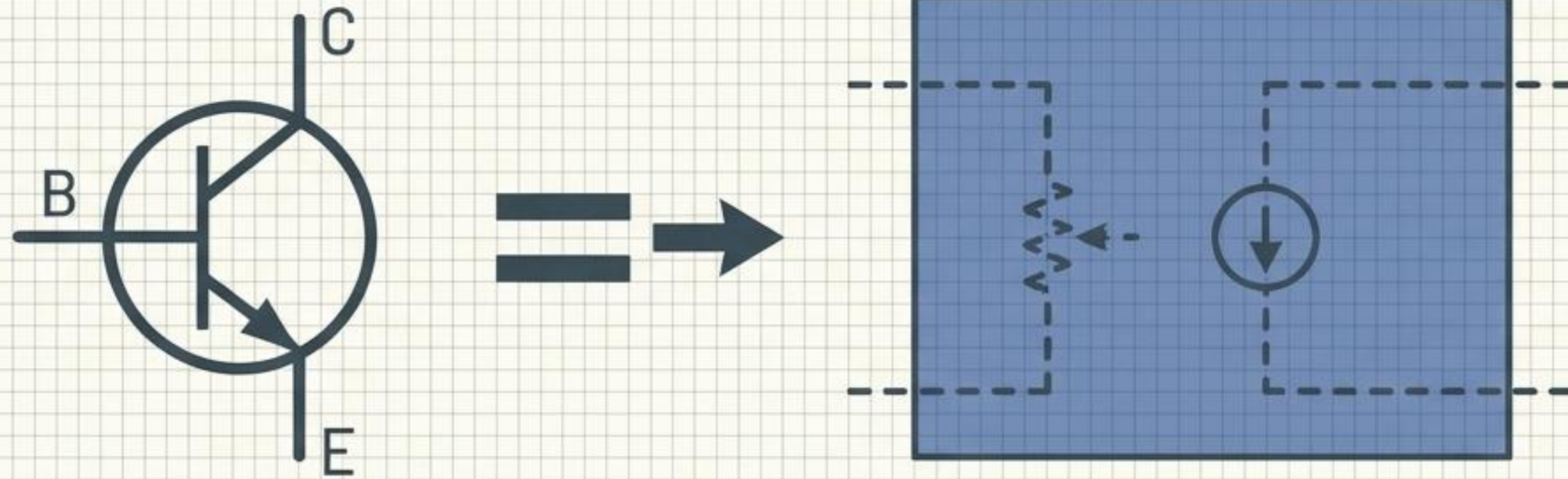
DC supply removed and grounded. Capacitors replaced with short-circuit equivalents.

The Clean Small-Signal Equivalent



The schematic redrawn for small-signal AC analysis. Notice how R_1 and R_2 are now in parallel to ground.

Inside the Black Box



- The physical transistor symbol is insufficient for mathematical calculation of dynamic AC signals.
- To calculate input impedance, output impedance, and voltage gain, engineers replace the physical transistor with an AC Equivalent Model.
- This model mathematically simulates exactly how the semiconductor junctions behave when a high-frequency AC signal passes through them.

The Modeling Landscape

The r_e Model

Focus: Built fundamentally around the internal AC emitter resistance (r_e'). Direct and highly practical for discrete amplifier design.

T and π Models

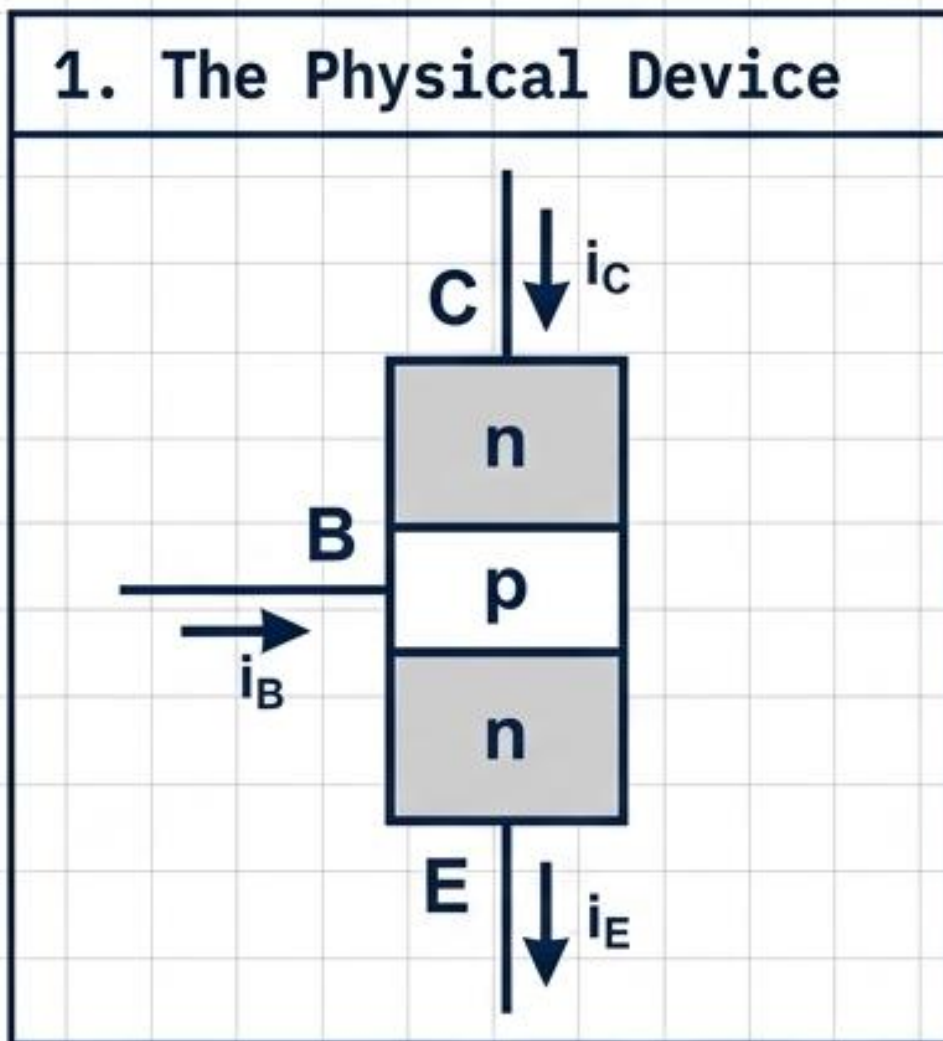
Focus: Widely utilized structural models specifically designed for small-signal AC analysis.

Excellent for nodal analysis and complex equivalent circuits.

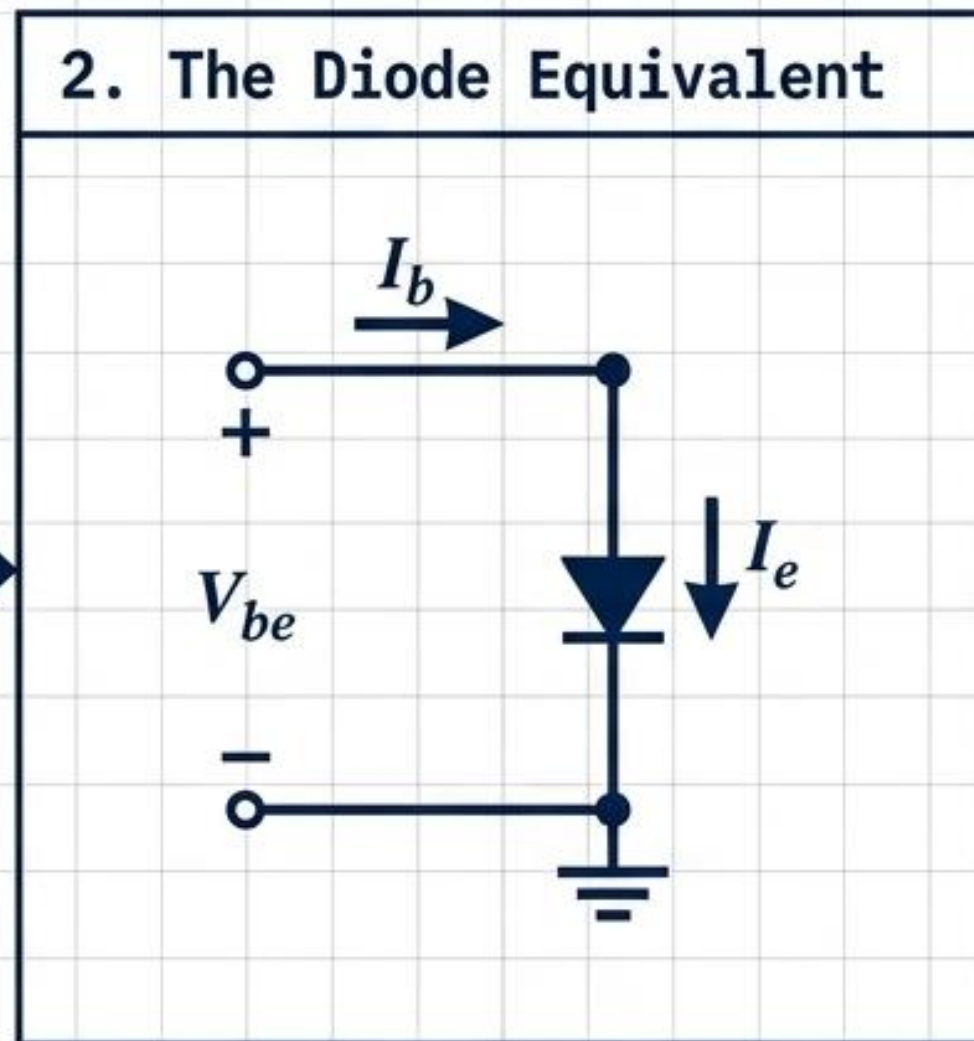
Hybrid Equivalent (h -parameter)

Focus: Utilizes specific hybrid parameters (h_{ie} , h_{fe} , etc.). This is the standard language used by semiconductor manufacturers on official BJT component datasheets.

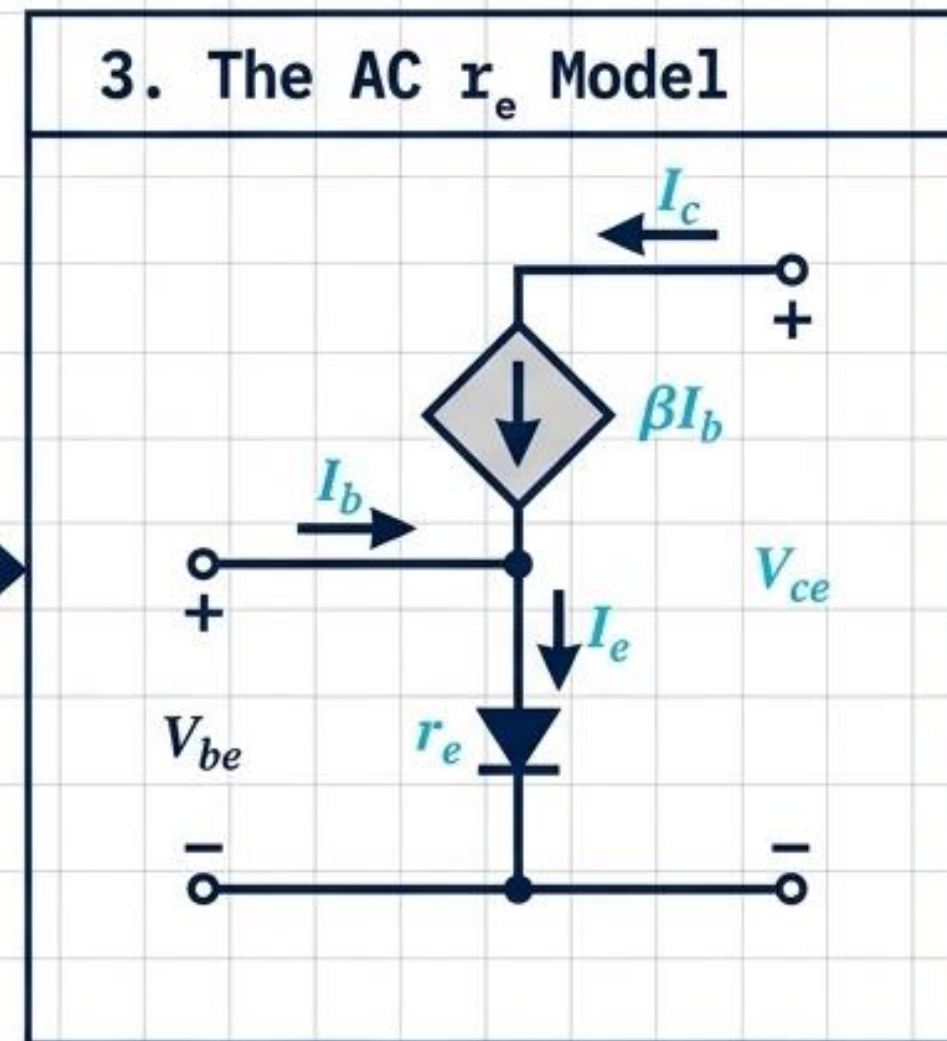
Conceptual Evolution: From Physics to AC Model



The BJT behaves like two physical diodes: an emitter-base (Forward Biased) and a collector-base (Reverse Biased).



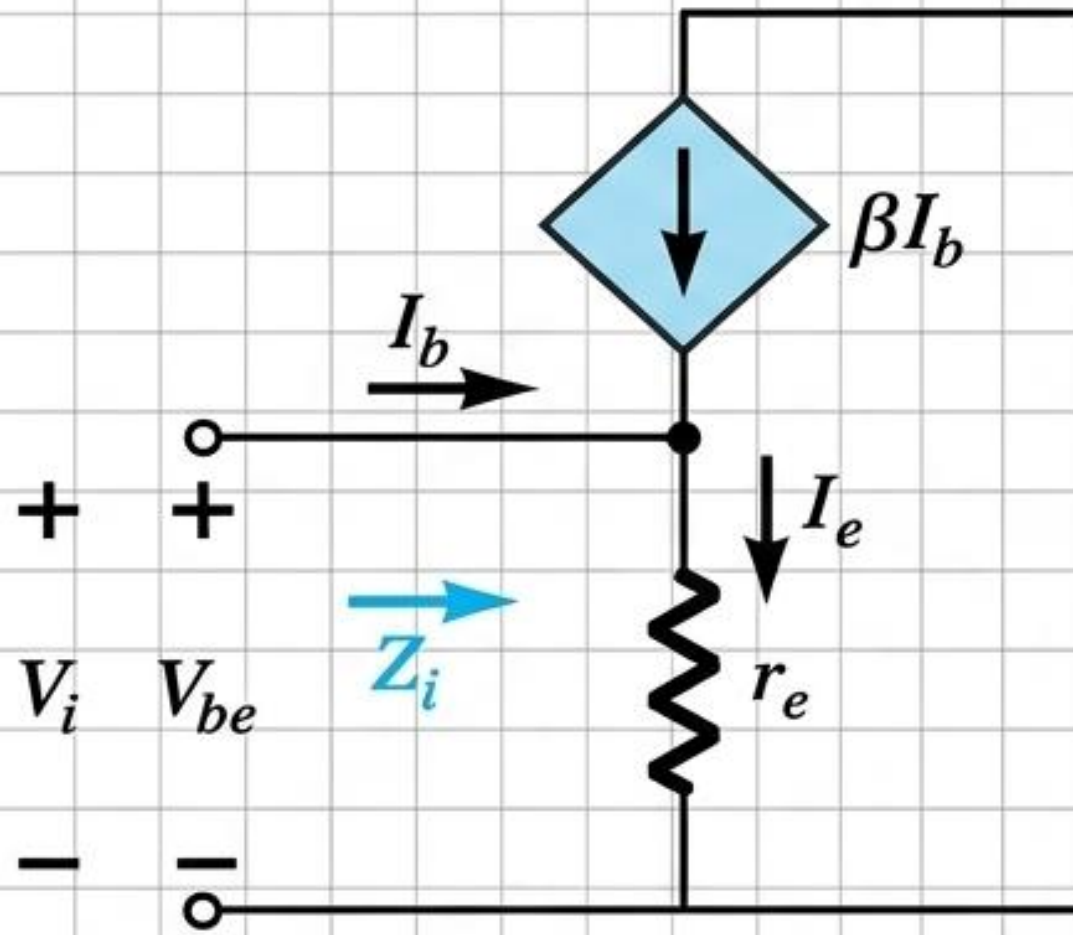
For the input equivalent circuit, the E-B junction is modeled as a simple forward-biased diode.



Under AC conditions, the diode is replaced by its dynamic resistance (r_e). The output side acts as a current-dependent current source.

$$i_c = \beta i_b \text{ (Output current} \\ = \beta \times \text{Input current)}$$

Bridging the Gap: Deriving the pi-Model



Question: Why is the input resistance looking into the base βr_e and not just r_e ?

Step 1: Express base-emitter voltage.

$$v_{be} = i_e \times r_e$$

Step 2: Relate emitter current to base current.

$$\text{Since } i_e \cong i_b \times \beta$$

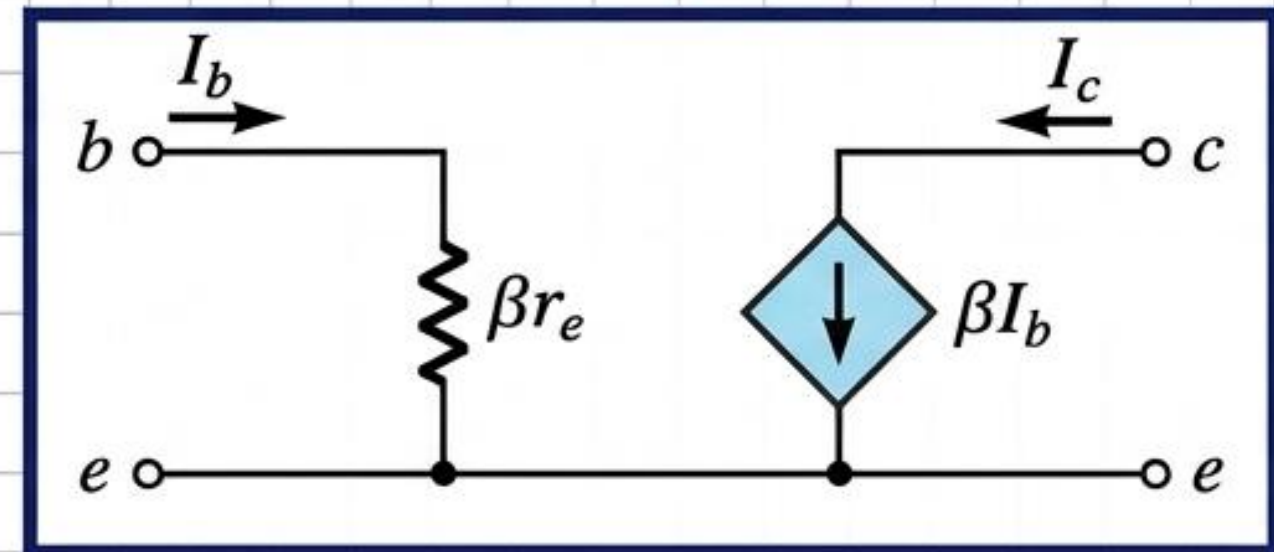
Step 3: Substitute i_e .

$$v_{be} \cong i_b \times \beta r_e$$

$$\text{More precisely: } v_{be} = (\beta + 1) i_b r_e$$

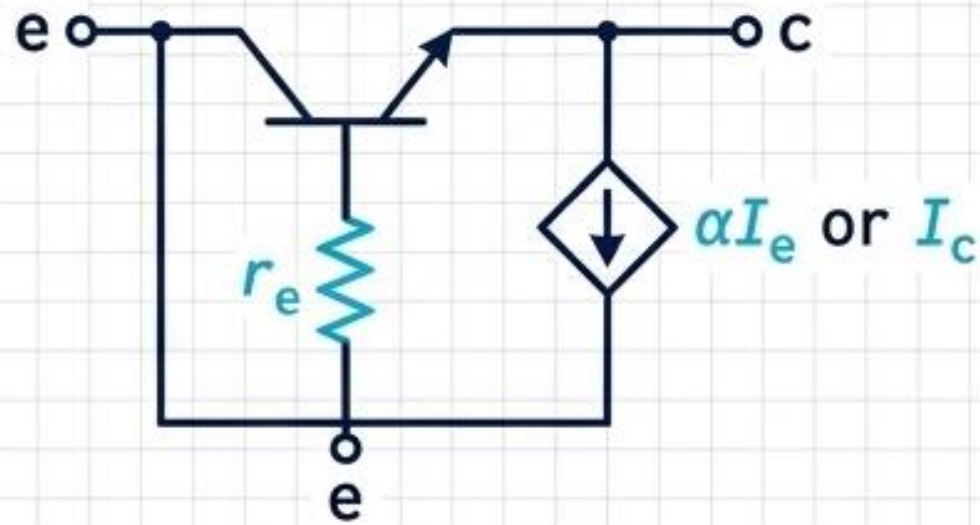
Conclusion:

$$\text{Therefore, } Z_{in(base)} = \beta r_e$$



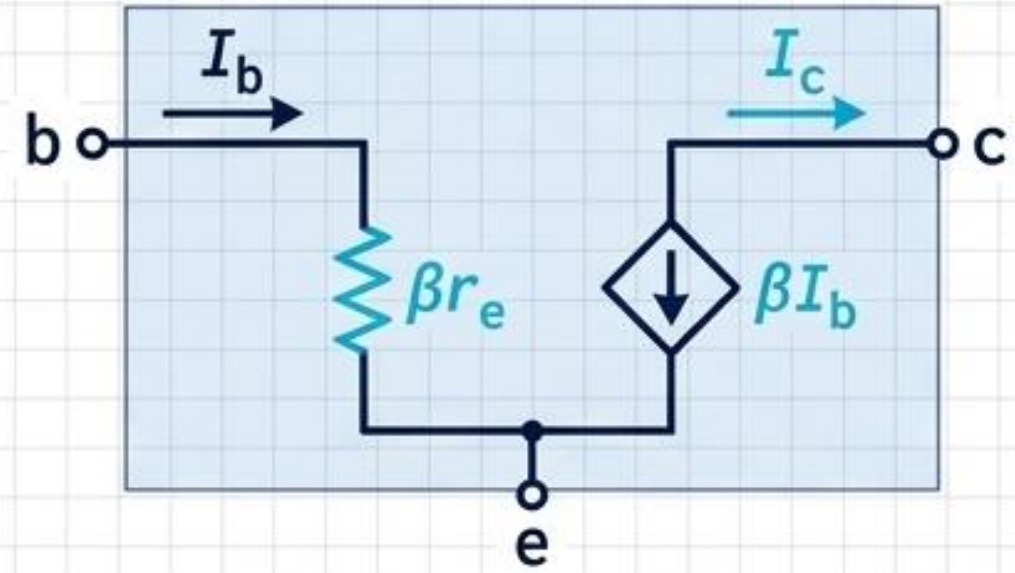
Overview of r_e Transistor Models

The T Model



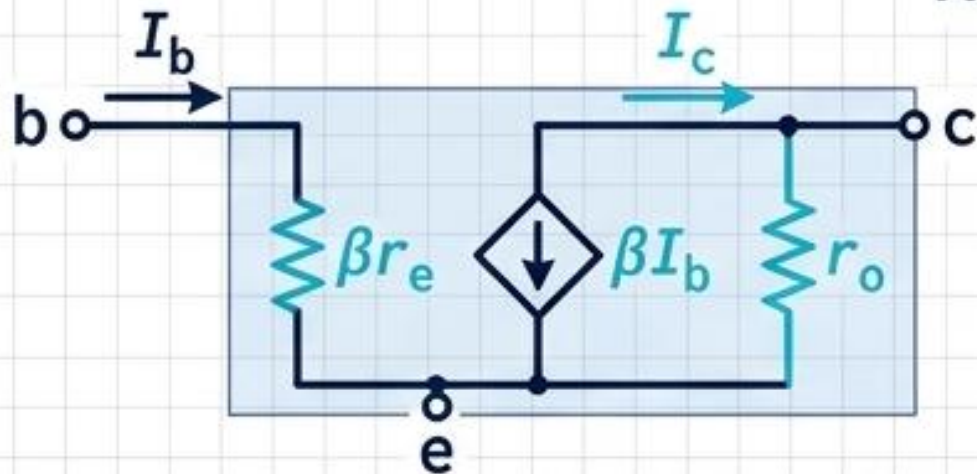
- Emphasizes the exact AC resistance (r_e) situated physically between the Emitter and Base.
- Best utilized for Common-Base configurations.

The π Model



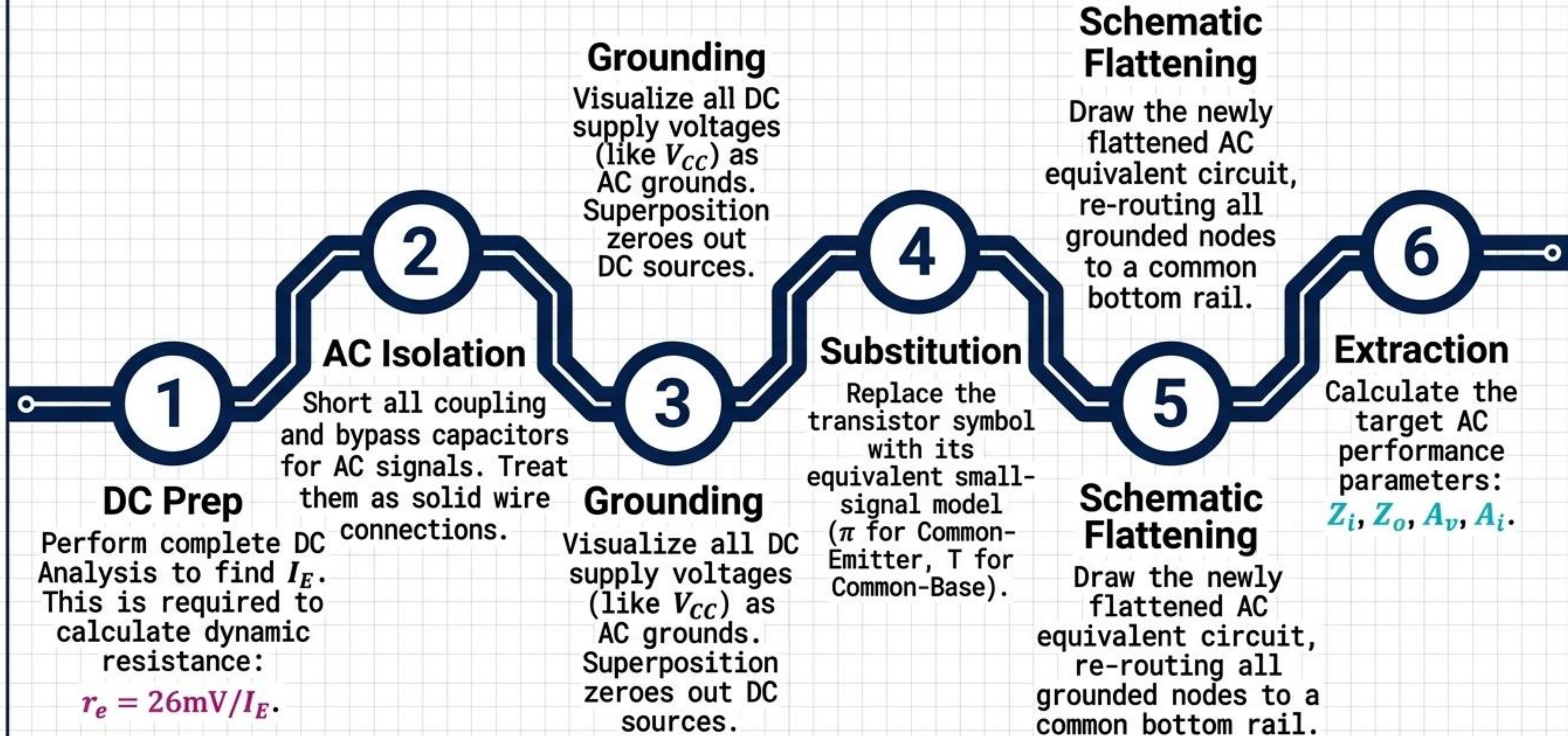
- Emphasizes the reflected impedance (βr_e) looking into the base.
- The absolute standard for Common-Emitter configurations.

The r_o Factor (Output Resistance)



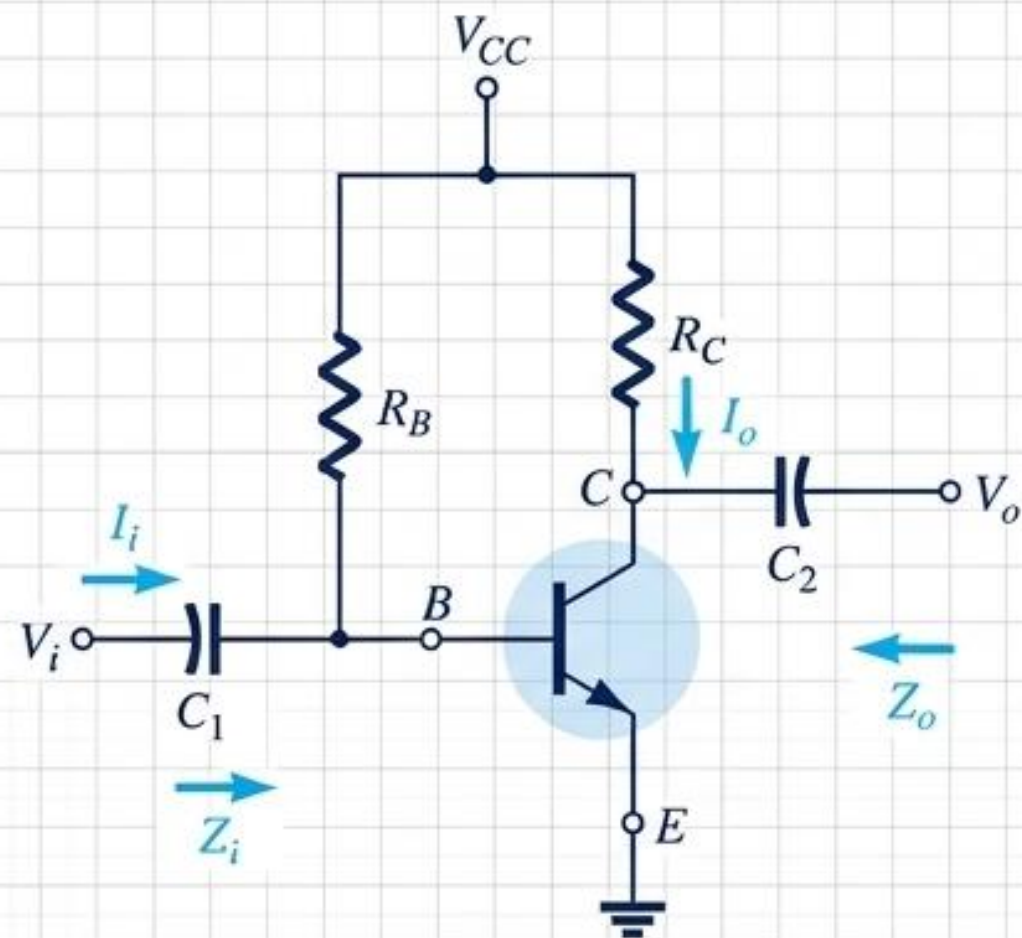
- In practical transistors, collector current depends slightly on collector voltage.
- Ideally: $r_o = \infty \Omega$ (Open circuit).
- Practically: It is finite (though very high) and placed in parallel with the output current source.

The Superposition Methodology (DC + AC Analysis)



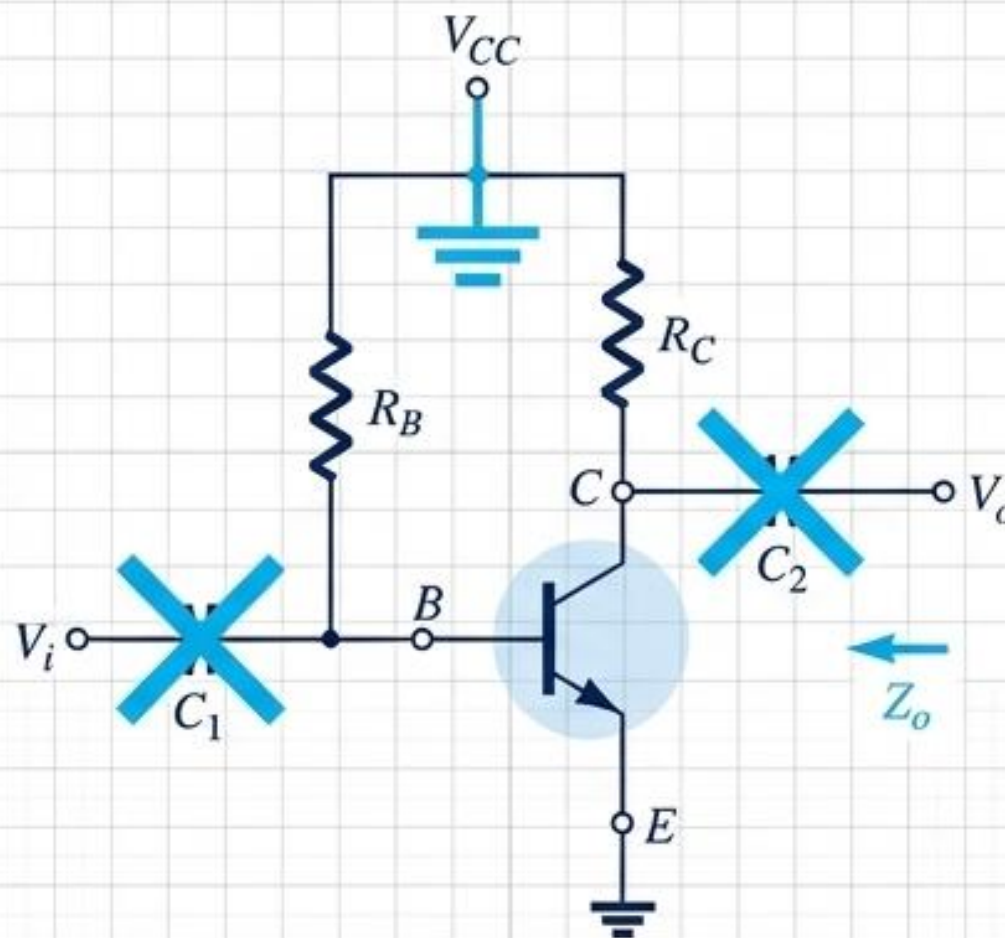
Topology 1: Common-Emitter Fixed-Bias Configuration

1. The Physical Circuit



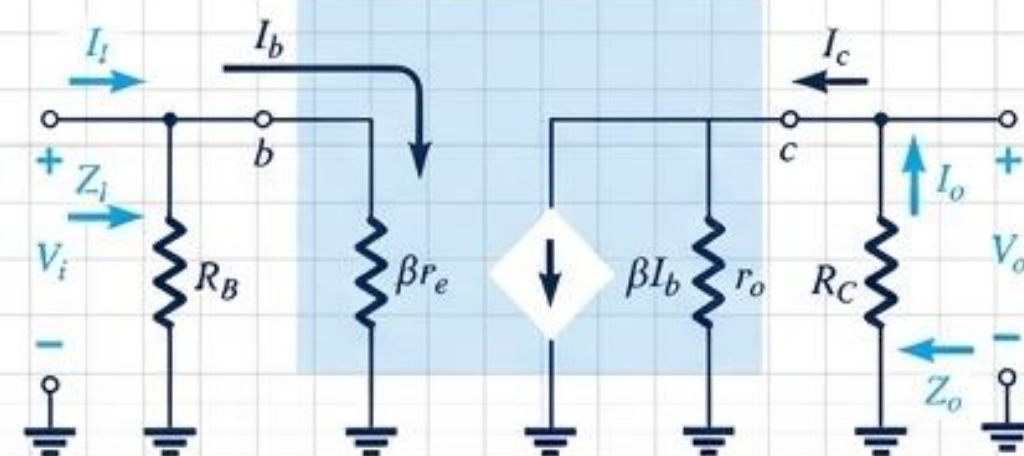
Original physical layout featuring coupling capacitors to block DC signals from input/output.

2. The Transition State



Apply methodology: Short capacitors for AC. Convert the V_{CC} DC supply to an AC ground.

3. The AC Equivalent



The flattened AC schematic. R_B is now in parallel with the pi-model input. Ready for mathematical extraction.

Fixed-Bias Analytics

Input Impedance

Z_i

$$Z_i = R_B \parallel \beta r_e$$

Condition: If $R_B \geq 10\beta r_e$, then $Z_i \cong \beta r_e$

Output Impedance

Z_o

$$Z_o = R_C \parallel r_o$$

Condition: If $r_o \geq 10 R_C$, then $Z_o \cong R_C$

Voltage Gain

A_v

$$A_v = -\frac{R_C \parallel r_o}{r_e}$$

Condition: If $r_o \geq 10 R_C$, then $A_v \cong -\frac{R_C}{r_e}$

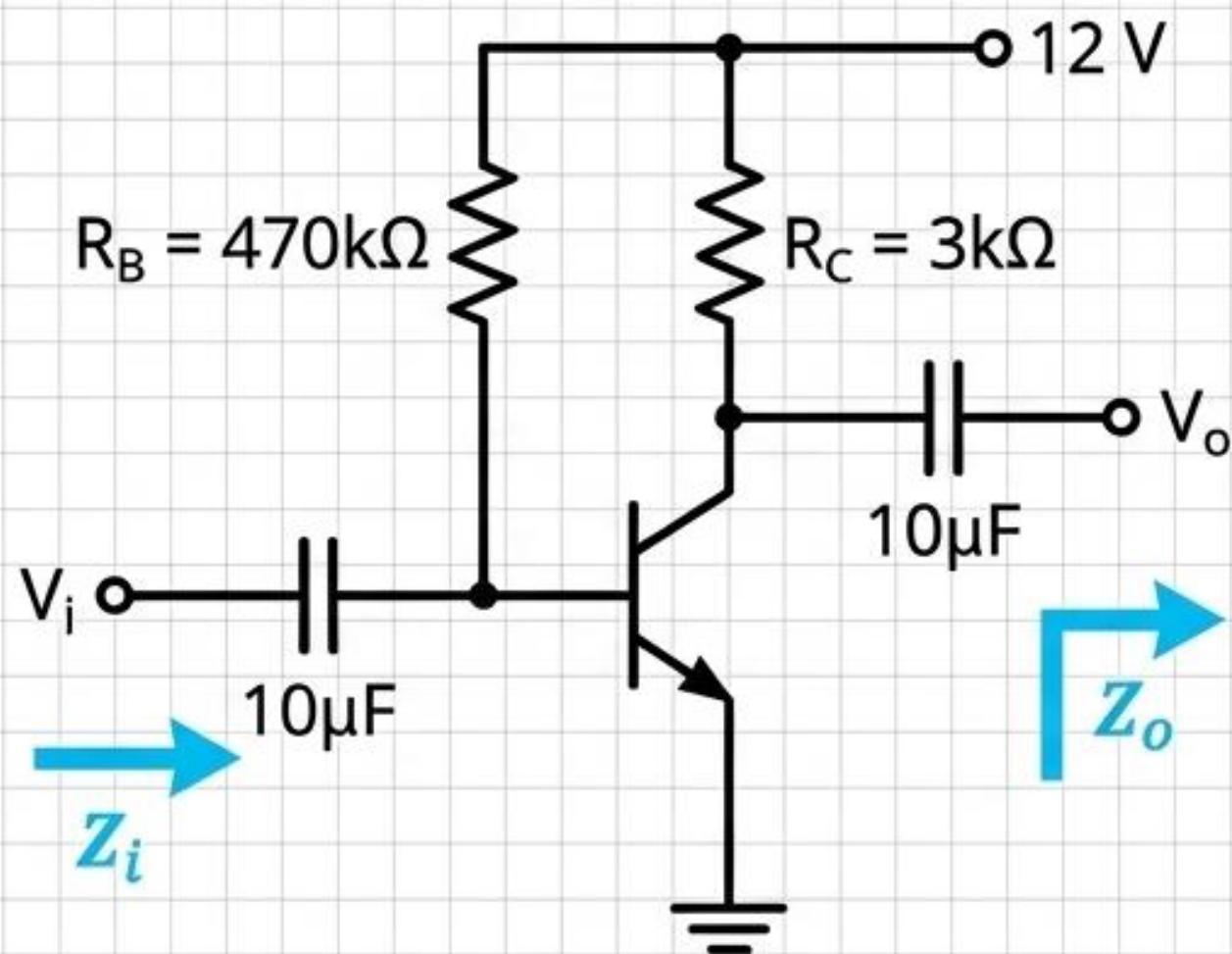
Current Gain

A_i

$$A_i = \frac{I_o}{I_i} = \frac{\beta R_B r_o}{(r_o + R_C)(R_B + \beta r_e)}$$

Condition: If $r_o \geq 10 R_C$ and $R_B \geq 10\beta r_e$, then $A_i \cong \beta$

Worked Example: Fixed-Bias Network



$$\beta = 100$$
$$r_o = 50\text{ k}\Omega$$

Phase 1: DC Analysis (Extracting r_e)

$$I_B = 24.04\text{ }\mu\text{A} \rightarrow I_E = 2.428\text{ mA}$$

$$\text{Result: } r_e = \frac{26\text{ mV}}{2.428\text{ mA}} = \mathbf{10.71\text{ }\Omega}$$

Phase 2: AC Parameters

$$\text{Pre-compute: } \beta r_e = (100)(10.71\text{ }\Omega) = 1.071\text{ k}\Omega$$

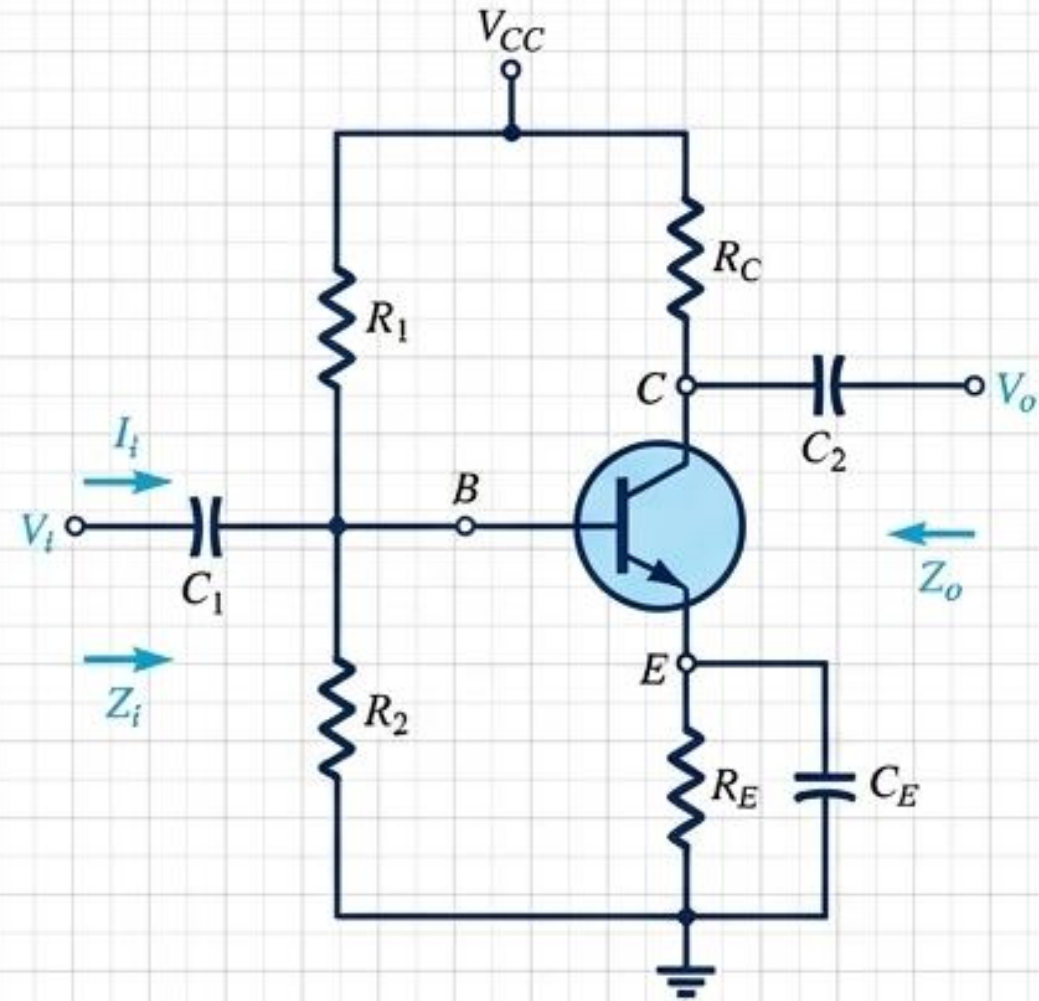
$$Z_i = 470\text{ k}\Omega \parallel 1.071\text{ k}\Omega = \mathbf{1.07\text{ k}\Omega}$$

$$Z_o = R_C = \mathbf{3\text{ k}\Omega}$$

$$A_v = -\frac{3\text{ k}\Omega}{10.71\text{ }\Omega} = \mathbf{-280.11}$$

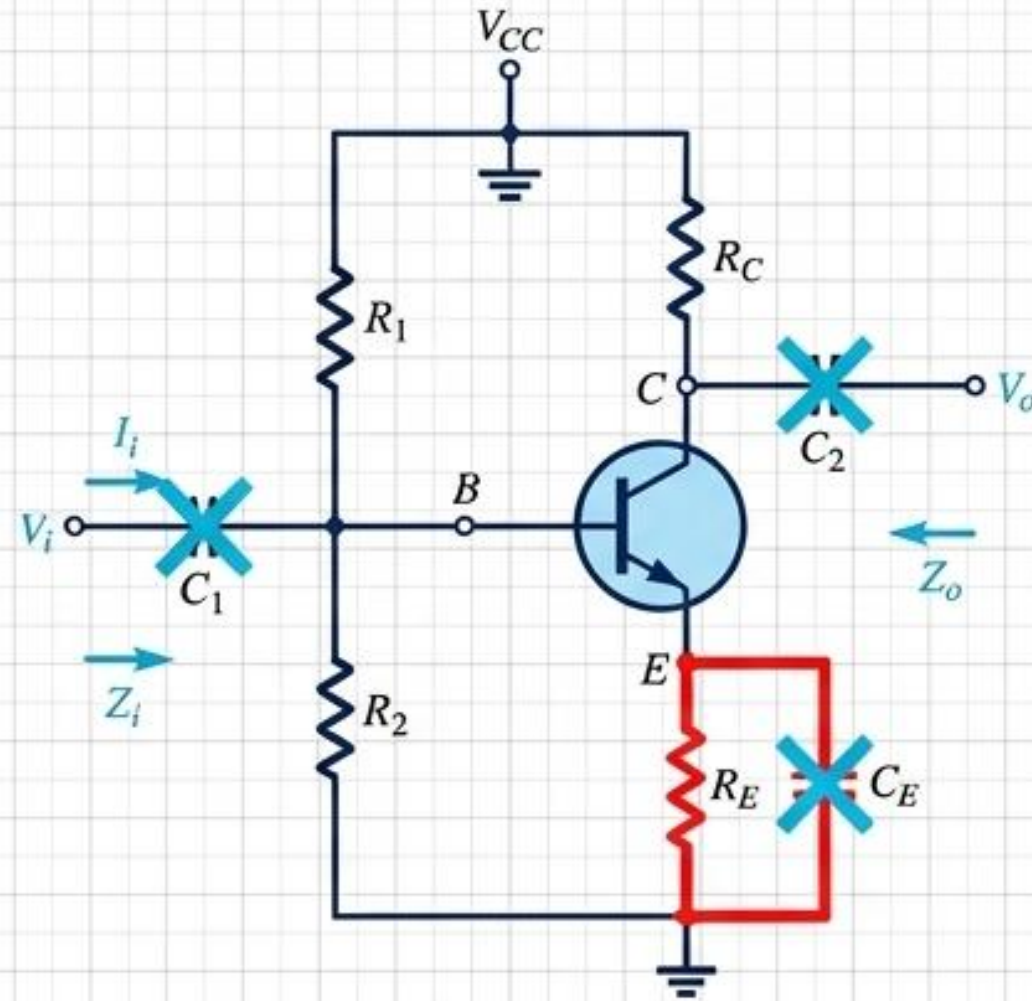
Topology 2: Common-Emitter Voltage-Divider Bias

1. The Physical Circuit



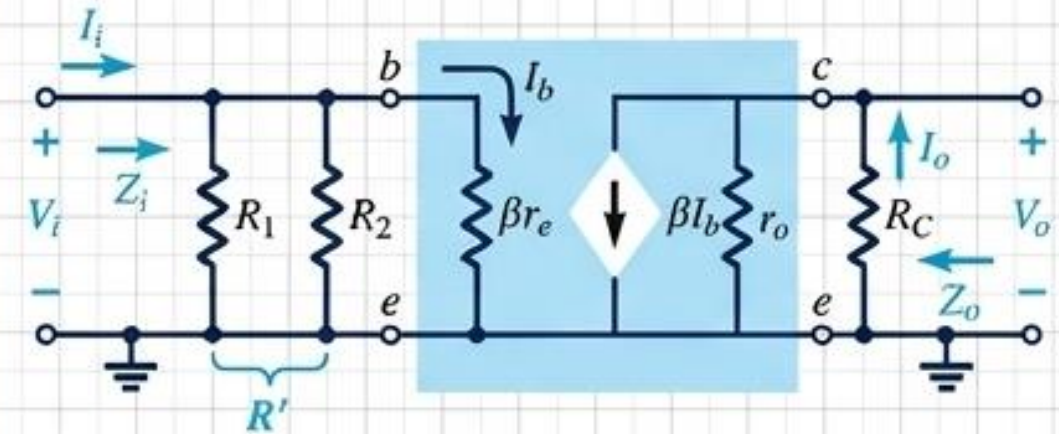
Original Voltage-Divider DC layout featuring the R_1/R_2 bias network and emitter bypass capacitor C_E .

2. The Transition State



Crucial step: Shorting C_E completely bypasses R_E , effectively connecting the Emitter directly to ground for AC analysis.

3. The AC Equivalent



The AC equivalent groups the parallel R_1 and R_2 into a single simplified input resistor denoted as R' .

Voltage-Divider Analytics

Pre-computation Definition: $R' = R_1 \parallel R_2$

Input Impedance

Z_i

$$Z_i = R' \parallel \beta r_e$$

Notice R_B is simply replaced by R' .

Output Impedance

Z_o

$$Z_o = R_C \parallel r_o$$

Condition: If $r_o \geq 10 R_C$, then $Z_o \cong R_C$

Voltage Gain

A_v

$$A_v = -\frac{R_C \parallel r_o}{r_e}$$

Condition: If $r_o \geq 10 R_C$, then $A_v \cong -\frac{R_C}{r_e}$

Current Gain

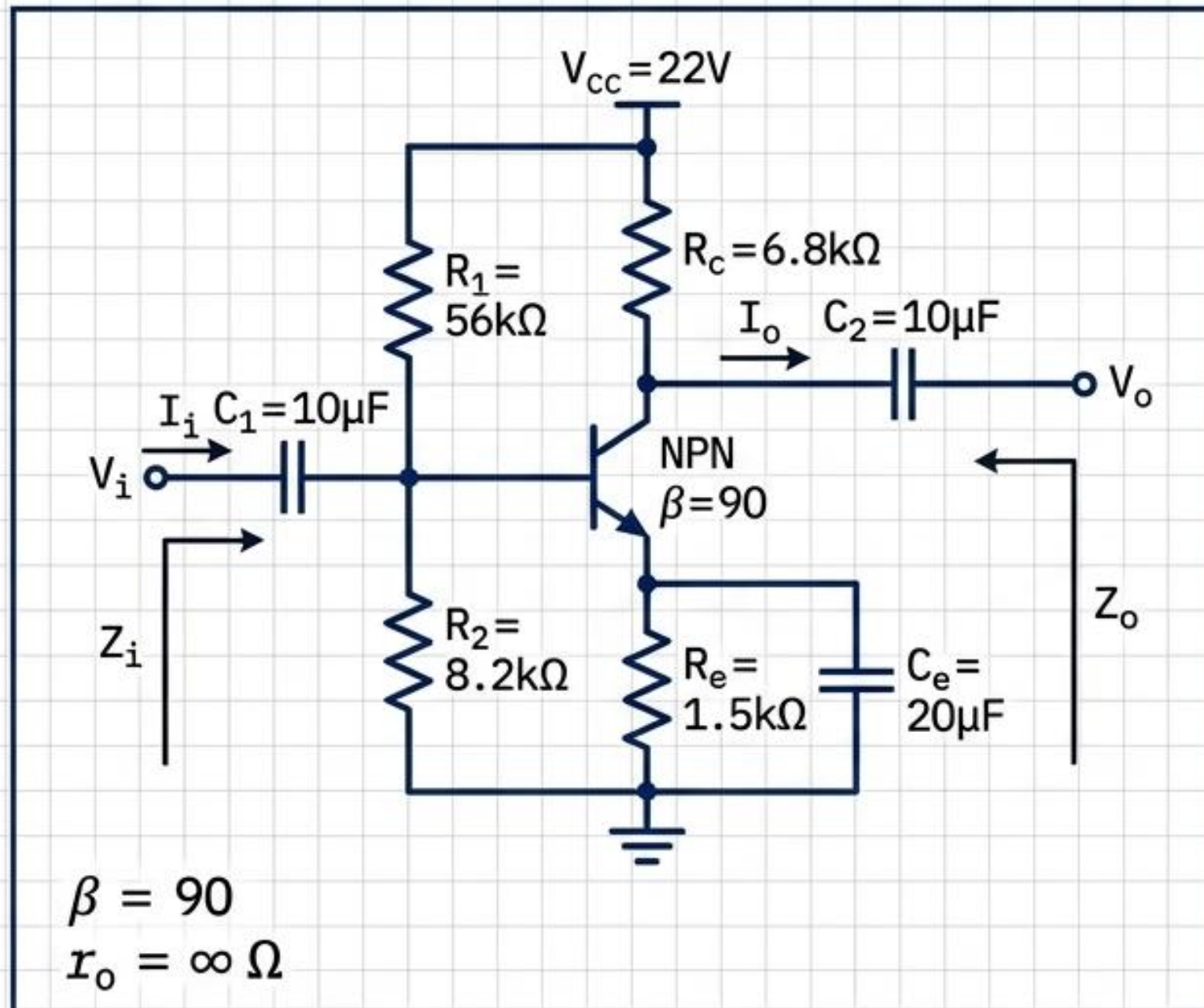
A_i

$$A_i = \frac{\beta R'}{R' + \beta r_e}$$

Assumes $r_o \geq 10 R_C$. If $R' \geq 10 \beta r_e$, then $A_i \cong \beta$.

Worked Example: Voltage-Divider Network

Schematic constraints



Calculation Waterfall

Phase 1: DC Analysis (Extracting r_e)

$$V_B = 2.81 \text{ V} \rightarrow V_E = 2.11 \text{ V} \rightarrow I_E = 1.41 \text{ mA}$$

$$\text{Result: } r_e = \frac{26 \text{ mV}}{1.41 \text{ mA}} = \mathbf{18.44 \Omega}$$

Phase 2: AC Parameters

$$R' = 56 \text{ k}\Omega \parallel 8.2 \text{ k}\Omega = 7.15 \text{ k}\Omega$$

$$Z_i = 7.15 \text{ k}\Omega \parallel (90)(18.44 \Omega) = \mathbf{1.35 \text{ k}\Omega}$$

$$Z_o = R_C = \mathbf{6.8 \text{ k}\Omega}$$

$$A_v = -\frac{6.8 \text{ k}\Omega}{18.44 \Omega} = \mathbf{-368.76}$$

Configuration Synthesis: Fixed-Bias vs. Voltage Divider

Dimension	Fixed-Bias	Voltage-Divider
AC Input Network	Relies on a single R_B resistor.	Relies on equivalent R' ($R_1 \parallel R_2$).
Input Impedance (Z_i)	$Z_i = R_B \parallel \beta r_e$	$Z_i = R' \parallel \beta r_e$ Typically yields a lower Z_i due to the parallel combination of R_1 and R_2 .
Voltage Gain (A_v)	$A_v \cong -R_C/r_e$ (Mathematical equivalent assuming bypassed emitter resistor).	$A_v \cong -R_C/r_e$
Key Differentiator: DC Stability	While AC parameter equations are nearly identical, the Voltage-Divider is heavily preferred in physical engineering practice. Its DC biasing point is largely independent of β , ensuring a mathematically stable r_e and reliable, repeatable AC performance across differing physical transistors.	